

EXPERIMENT: 8

A PYTHON PROGRAM TO IMPLEMENT BOOSTING

Aim:

To implement Boosting using AdaBoost algorithm.

Procedure:

1. Import required libraries.
2. Load dataset (Breast Cancer dataset).
3. Split dataset into training and testing sets.
4. Train AdaBoost classifier.
5. Predict output for test data.
6. Calculate accuracy.

CODE:

```
from sklearn.datasets import load_wine

from sklearn.model_selection import train_test_split

from sklearn.ensemble import AdaBoostClassifier

from sklearn.metrics import accuracy_score


data = load_wine()


X = data.data

y = data.target
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
model = AdaBoostClassifier(n_estimators=50)  
  
model.fit(X_train, y_train)  
  
y_pred = model.predict(X_test)  
  
print("Accuracy:", accuracy_score(y_test, y_pred))
```

OUTPUT:

Accuracy: 0.9444

RESULT:

The AdaBoost model improved classification accuracy by combining multiple weak learners